ORIGINAL ARTICLE

CRISPR-Based Therapy for Hereditary Angioedema

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ABSTRACT

BACKGROUND

Hereditary angioedema is a rare genetic disease characterized by severe and unpredictable swelling attacks. NTLA-2002 is an in vivo gene-editing therapy that is based on clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated protein 9. NTLA-2002 targets the gene encoding kallikrein B1 (*KLKB1*). A single dose of NTLA-2002 may provide lifelong control of angioedema attacks.

METHODS

In this phase 2 portion of a phase 1–2 trial, we randomly assigned adults with hereditary angioedema in a 2:2:1 ratio to receive NTLA-2002 in a single dose of 25 mg or 50 mg or placebo. The primary end point was the number of angioedema attacks per month (the monthly attack rate) from week 1 through week 16. Secondary end points included safety, pharmacokinetics, and pharmacodynamics (i.e., the change from baseline in total plasma kallikrein protein level); exploratory end points included patient-reported outcomes.

RESULTS

Of the 27 patients who underwent randomization, 10 received 25 mg of NTLA-2002, 11 received 50 mg, and 6 received placebo. From week 1 through week 16, the estimated mean monthly attack rate was 0.70 (95% confidence interval [CI], 0.25 to 1.98) with 25 mg of NTLA-2002, 0.65 (95% CI, 0.24 to 1.76) with 50 mg, and 2.82 (95% CI, 0.80 to 9.89) with placebo; the difference in the estimated mean attack rate with NTLA-2002 as compared with placebo was −75% with 25 mg and −77% with 50 mg. Among patients who received NTLA-2002, 4 of the 10 patients who received 25 mg (40%) and 8 of the 11 who received 50 mg (73%) were attack-free with no additional treatment during the period from week 1 through week 16. The most common adverse events among patients who received NTLA-2002 were headache, fatigue, and nasopharyngitis. The mean percent change in total plasma kallikrein protein levels from baseline to week 16 was −55% with 25 mg and −86% with 50 mg; levels remained unchanged with placebo.

CONCLUSIONS

NTLA-2002 administered in a single dose of 25 mg or 50 mg reduced angioedema attacks and led to robust and sustained reduction in total plasma kallikrein levels in patients with hereditary angioedema. These results support continued investigation in a larger phase 3 trial. (Funded by Intellia Therapeutics; ClinicalTrials.gov number, NCT05120830; EudraCT number, 2021-001693-33.)

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A complete list of subprincipal investigators and study-site coordinators in this trial is provided in the Supplementary Appendix, available at NEJM.org.

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Hereditary angioedema is a rare, autosomal dominant disorder that affects approximately 1 in 50,000 persons worldwide. It is a debilitating and potentially fatal condition that manifests as recurrent episodes of unpredictable swelling.[1,2] Disease onset often occurs in childhood, with attacks recurring throughout a person's lifetime at intervals of every few days to weeks and commonly lasting 3 to 5 days when untreated.[2] The most frequently affected organs are the gastrointestinal tract and the cutaneous tissues of the body; effects on the former are most often associated with colicky pain, and effects on the latter can be disabling or disfiguring.[1-5] Furthermore, laryngeal edema can lead to life-threatening airway obstruction and death by asphyxiation.[1,2]

Hereditary angioedema is caused predominantly by a number of variants in the gene *SERPING1*. This gene encodes the protein C1 esterase inhibitor (C1INH), a key regulator in the contact activation pathway and in bradykinin production through inhibition of factor XIIa and plasma kallikrein. These variants lead to either C1INH deficiency (in type 1 disease) or C1INH dysfunction (in type 2 disease), either of which results in excessive levels of bradykinin, which in turn lead to episodes of increased vascular hyperpermeability and subsequent swelling.[1,2,6]

Plasma kallikrein is clinically validated as a therapeutic target, with kallikrein inhibitors approved for long-term prophylaxis.[1,7-10] RNA-silencing therapies that reduce plasma kallikrein protein levels have also shown efficacy in clinical trials.[11-13] Inhibition or reduction of plasma kallikrein reduces the amount of bradykinin.[8] Although new prophylactic treatments have decreased the frequency and severity of attacks and improved the lives of persons with hereditary angioedema in recent years, these treatments require regular and lifelong administration, which affects overall quality of life.[2,8,9,14-16]

NTLA-2002 is a nonviral gene-editing therapy that is based on clustered regularly interspaced short palindromic repeats (CRISPR)–CRISPR-associated protein 9 (Cas9); it is delivered in a single dose, administered intravenously.[17] (CRISPR-Cas9 is also known, and more accurately described, as guide RNA–Cas9.) NTLA-2002 is designed to permanently edit the gene encoding kallikrein B1, *KLKB1*, in hepatocytes, the primary source of plasma prekallikrein; this editing results in a reduction in the total plasma kallikrein protein level, which should prevent angioedema attacks.[17-21]

Previously reported results from the phase 1 dose-escalation portion of this phase 1–2 trial showed no dose-limiting toxic effects from a single infusion of NTLA-2002 at a dose of 25 mg, 50 mg, or 75 mg. Patients had a sustained, dose-dependent reduction in plasma kallikrein protein levels and a 95% reduction from baseline in angioedema attacks.[17] Here, we report results from the phase 2 portion of the phase 1–2 trial and additional follow-up results from phase 1.

Methods

Trial Design and Oversight

We conducted the phase 2 portion of this ongoing, phase 1–2, double-blind, randomized, placebo-controlled trial to evaluate the efficacy, safety, pharmacodynamics, and pharmacokinetics of NTLA-2002 at doses of 25 mg and 50 mg in adults with hereditary angioedema. Patients were enrolled at nine sites in Australia, France, Germany, the Netherlands, New Zealand, and the United Kingdom. The trial was approved by the institutional review board or ethics committee at each participating trial site. All patients provided written informed consent. The trial was performed in accordance with the principles of the Declaration of Helsinki and the current Good Clinical Practice guidelines of the International Council for Harmonisation. An independent data monitoring committee provided trial oversight. The contributions of individual authors are listed in the Supplementary Appendix, available with the full text of this article at NEJM.org. The trial was conceived and designed by three authors representing the sponsor and one academic author. Data were collected by authors from academic institutions and representatives of the sponsor. Representatives of the sponsor analyzed the data. The first draft of the manuscript was written by a representative of the sponsor and sponsor-funded medical writers. All the authors contributed to data interpretation, participated in reviewing and editing earlier versions of the manuscript, agreed to submit the manuscript for publication, and approved the submitted version of the manuscript. The authors vouch for the completeness and accuracy of the data and for the fidelity of the trial to the protocol (available at NEJM.org) and to the statistical analysis plan described in the protocol. All the authors had access to all data in the trial and agreed to maintain confidentiality of the data during manuscript development.

PATIENTS

Eligible patients were at least 18 years of age, had received a diagnosis of type 1 or type 2 hereditary angioedema, and had had three or more investigator-confirmed angioedema attacks within 90 days before the screening period and two or more investigator-confirmed attacks during the 8-week screening period. Patients were not permitted to use prophylactic therapies within five half-lives before the beginning of the screening period through the end of the 16-week primary observation period. Full eligibility criteria are listed in the protocol.

RANDOMIZATION AND TREATMENT

Patients were randomly assigned in a 2:2:1 ratio to receive NTLA-2002 administered intravenously in a single dose of 25 mg or 50 mg or a placebo saline solution. Dose levels of 25 mg and 50 mg were selected on the basis of results from the phase 1 study.[17] On day 1, NTLA-2002 or placebo was administered by infusion over the course of 2 to 6 hours. To mitigate the risk of infusion-related reactions, patients followed a pretreatment regimen on the day before and the day of infusion, as described previously[17] and in the protocol. In brief, 8 to 24 hours before infusion, an oral dose of a glucocorticoid was administered; 1 to 2 hours before infusion, prophylactic treatment consisting of a glucocorticoid, an H_1 blocker, and an H_2 blocker was administered. C1INH therapy was required to be available at the bedside during the infusion in case of unexpected onset of a swelling attack. All patients were permitted to use on-demand medication for angioedema attacks at their own discretion at any time during the trial; permitted and prohibited concomitant medications are listed in the Supplementary Appendix. The primary observation period was the first 16 weeks after infusion, and this period was followed by a long-term observation period of 88 weeks.

CLINICAL ASSESSMENT

Each patient recorded angioedema symptoms, attacks, and treatments in an electronic diary. An investigator reviewed the diary at each trial visit to confirm whether an event represented an attack due to angioedema. Blood and urine samples for pharmacodynamic and pharmacokinetic analyses were obtained at each trial visit in accordance with the schedule of activities. Adverse events were graded according to the National Cancer Institute Common Terminology Criteria for Adverse Events, version 5.0.[22] Additional tests and evaluations performed at each visit are described in the protocol.

TRIAL END POINTS

The primary end point was the number of angioedema attacks per month (the monthly attack rate) during the primary observation period, with a month defined as 28 days. Secondary end points included safety (assessed on the basis of adverse events, which were defined as any events that began or worsened after the administration of NTLA-2002 or placebo), the number of angioedema attacks per month from week 5 through week 16, the number of angioedema attacks per month from week 1 through week 16 and from week 5 through week 16 that resulted in the use of on-demand therapy, the number of moderate or severe angioedema attacks per month from week 5 through week 16, the change from baseline in the total plasma kallikrein protein level, and plasma concentrations of the components of NTLA-2002 (exogenous lipids [LP000001 and DMG-PEG2k], Cas9 messenger RNA, and *KLKB1* single guide RNA). Exploratory end points included the change from baseline in the use of on-demand medication for angioedema attacks, the change from baseline in plasma kallikrein protein activity, and the change from baseline in quality-of-life variables assessed by the Moxie Angioedema Quality of Life questionnaire (AE-QoL) and other measures as described in the protocol. The AE-QoL is a validated, angioedema-specific assessment of patient-reported outcomes. Scores for each of the four individual domains and total scores range from 0 to 100, with lower scores indicating less impairment; the minimal clinically important difference is defined as a change of 6 points in the AE-QoL total score.[23]

STATISTICAL ANALYSIS

We planned to enroll a total of up to 25 patients (20 patients in the NTLA-2002 groups and 5 in the placebo group) in the phase 2 portion of the trial. The sample size was not determined by a formal power analysis but was selected to provide an adequate number of patients for overall evaluation of the efficacy, safety, pharmacodynamics, and pharmacokinetics of NTLA-2002 at doses of 25 mg and 50 mg. No formal statistical testing was planned. The primary analysis was performed when the 25th patient reached week 16. We analyzed the primary end point using a negative binomial regression model, adjusting for the baseline monthly attack rate. The least-squares mean monthly attack rate and the corresponding 95% confidence interval for each trial group, as well as the mean ratio of the monthly attack rate with NTLA-2002 relative to that with placebo and the corresponding 95% confidence interval, were estimated with this model. The 95% confidence intervals presented have not been adjusted for multiplicity and should not be used in place of hypothesis testing.

The full analysis population, which was used for efficacy analyses, included all patients who underwent randomization. The safety analysis population included all patients who underwent randomization and received at least a partial dose of NTLA-2002 or placebo. Additional information regarding the statistical analysis is provided in the Supplementary Appendix.

RESULTS

PATIENTS

Of 36 patients screened, 27 underwent randomization: 10 patients were assigned to receive 25 mg of NTLA-2002, 11 were assigned to receive 50 mg, and 6 were assigned to receive placebo. Seven patients were ineligible according to prespecified criteria, and 2 withdrew consent before enrollment (Fig. S1 in the Supplementary Appendix). As of April 4, 2024 (the data-cutoff date), the median follow-up time was 8.2 months (range, 4.4 to 11.8) in the 25-mg group, 5.6 months (range, 2.9 to 11.5) in the 50-mg group, and 6.9 months (range, 1.9 to 12.5) in the placebo group. At the time of analysis, 2 patients had not had a week 16 assessment; however, all patients were included in the primary and follow-up analyses. The baseline characteristics of patients in the NTLA-2002 and placebo groups were similar (Table 1). Table S1 provides information on the representativeness of the patient population studied. The median age was 48 years (range, 18 to 61) in the NTLA-2002 groups and 47 years (range, 31 to 76) in the placebo group; 57% of patients in the NTLA-2002 groups, as compared with 33% in the placebo group, were men.

EFFICACY

The mean monthly attack rate at baseline was 3.6 in both the 25-mg and 50-mg groups and 3.7 in the placebo group. During the 16-week primary observation period, the estimated mean monthly attack rate (adjusted for the baseline value) was 0.70 (95% confidence interval [CI], 0.25 to 1.98) in the 25-mg group, 0.65 (95% CI, 0.24 to 1.76) in the 50-mg group, and 2.82 (95% CI, 0.80 to 9.89) in the placebo group. The mean difference in the monthly attack rate with NTLA-2002 as compared with placebo was −75% (95% CI, −95 to 27) with 25 mg and −77% (95% CI, −95 to 15) with 50 mg (Table 2, Fig. 1, and Fig. S2). Similar results were observed for the monthly attack rate from week 5 through week 16. The mean (±SD) percent change from baseline in the monthly attack rate from week 1 through week 16 was −78.1±30.4% with 25 mg of NTLA-2002, −79.5±42.0% with 50 mg, and −16.3±41.6% with placebo (Table S2). Among patients who received NTLA-2002, four (40%) in the 25-mg group and eight (73%) in the 50-mg group were attack-free and received no other treatment between week 1 and week 16. No patients in the placebo group were free of attacks during this period (Fig. S3). All eight patients in the 50-mg group who were attack-free during the primary observation period remained attack-free as of the latest assessment. The estimated mean difference between the NTLA-2002 groups and the placebo group in the number of angioedema attacks per month that resulted in on-demand therapy during week 1 through week 16 was −74% (95% CI, −95 to 51) with 25 mg of NTLA-2002 and −77% (95% CI, −96 to 39) with 50 mg; the estimated mean difference in the number of moderate-to-severe attacks per month with NTLA-2002 as compared with placebo was −78% (95% CI, −96 to 26) with 25 mg and −85% (95% CI, −97 to −11) with 50 mg (Table S3). After week 16, four patients who had received placebo and two patients who had received NTLA-2002 (one in each dose group) resumed long-term prophylaxis (Fig. 1).

Table 1. Characteristics of the Patients at Baseline.

Characteristic	NTLA-2002, 25 mg (N=10)	NTLA-2002, 50 mg (N=11)	Placebo (N=6)
Median age (range) — yr	48.5 (34–61)	44.0 (18–61)	47.0 (31–76)
Male sex — no. (%)	7 (70)	5 (45)	2 (33)
Race or ethnic group — no. (%)*			
White	9 (90)	11 (100)	6 (100)
Pacific Islander	1 (10)	0	0
Median weight (range) — kg	89 (58–133)	78 (56–107)	79 (50–92)
Type of hereditary angioedema — no. (%)			
C1INH type 1	8 (80)	10 (91)	5 (83)
C1INH type 2	2 (20)	1 (9)	1 (17)
Previous receipt of long-term prophylaxis — no. (%)	6 (60)	6 (55)	5 (83)
Long-term prophylaxis before trial enrollment — no. (%)			
Lanadelumab	2 (20)	2 (18)	1 (17)
Attenuated androgens	2 (20)	1 (9)	2 (33)
Berotralstat	1 (10)	2 (18)	1 (17)
C1INH	1 (10)	0	1 (16.7)
Tranexamic acid	0	1 (9)	0
Typical severity of angioedema attacks — no. (%)			
Mild	1 (10)	0	1 (17)
Moderate	6 (60)	9 (82)	4 (67)
Severe	3 (30)	2 (18)	1 (17)
Previous occurrence of laryngeal attacks — no. (%)†	6 (60)	5 (45)	4 (67)
Median no. of angioedema attacks during historical attack period (range)‡	6.5 (3–24)	4.0 (3–11)	5.5 (3–9)
Type of angioedema attack — no. (%)†			
Peripheral	8 (80)	9 (82)	6 (100)
Abdominal	9 (90)	9 (82)	6 (100)
Laryngeal	6 (60)	5 (45)	4 (67)

* Race or ethnic group was reported by the patients.
† Included are patients who had the indicated type of attack at any point in their medical history.
‡ The historical attack period is defined as the 90 days before the screening period, which coincided with the washout of any long-term prophylaxis medications taken by patients before trial enrollment.

SAFETY

All patients received the intended dose of NTLA-2002 or placebo. The most common adverse events (those that occurred in ≥25% of patients) in the two NTLA-2002 groups combined were headache (in 38% of patients), fatigue (in 29%), and nasopharyngitis (in 29%) (Table 3). All adverse events were grade 1 or 2 except for one event that occurred in a patient in the placebo group who had a serious adverse event of grade 4 edema of the tongue with breathing impairment, which was attributed to the underlying hereditary angioedema. No other serious adverse events occurred. Grade 2 infusion-related reactions were reported by two patients receiving NTLA-2002 (one in each treatment group), which led to temporary interruption of the infusion. In each instance, the reaction resolved without sequelae, and both patients received the full dose with the use of reduced infusion rates. All other infusion-related reactions were grade 1 (Table S4); symptoms of infusion-related reactions

Table 2. Mean Number of Investigator-Confirmed Angioedema Attacks per Month.*

Variable	NTLA-2002, 25 mg (N=10)	NTLA-2002, 50 mg (N=11)	Placebo (N=6)
Weeks 1 through 16			
Mean no. of attacks per month (95% CI)	0.70 (0.25–1.98)	0.65 (0.24–1.76)	2.82 (0.80–9.89)
Percent difference vs. placebo (95% CI)	−75 (−95 to 27)	−77 (−95 to 15)	
Weeks 5 through 16			
Mean no. of attacks per month (95% CI)	0.64 (0.21–1.94)	0.59 (0.20–1.71)	3.14 (0.84–11.80)
Percent difference vs. placebo (95% CI)	−80 (−96 to 14)	−81 (−97 to 3)	

* The mean number of angioedema attacks per month was estimated with the use of a negative binomial model with the trial group and the number of attacks per month at baseline as independent variables. Baseline is defined as the period from the date of informed consent to randomization. A month is defined as 28 days.

that were reported included back pain, flushing, noncardiac chest pain, generalized tingling, neck pain, and tight chest. One patient who received 25 mg of NTLA-2002 had an asymptomatic grade 2 increase in the alanine aminotransferase level on day 22, which was considered by the trial site investigator to be possibly related to treatment with NTLA-2002. The patient's alanine aminotransferase level decreased to a grade 1 level by day 28 and returned to normal by week 8. No clinically significant laboratory results were recorded for any patients.

Pharmacodynamics and Pharmacokinetics

Total plasma kallikrein protein levels were reduced from baseline levels in a dose-dependent manner in patients who received NTLA-2002, with a mean percent change from baseline to week 16 of −55±23% with 25 mg and −86±6.9% with 50 mg; levels remained unchanged in the placebo group (Fig. 2). Pharmacokinetic data are summarized in Figure S4.

Patient-Reported Outcomes

Patients in the NTLA-2002 groups had greater improvement from baseline in the AE-QoL total score at week 16 than those in the placebo group, with a mean change from baseline of −10.8±10.9 in the 25-mg group, −18.2±17.2 in the 50-mg group, and −3.5±10.4 in the placebo group (Fig. S5). Among the four domains of the AE-QoL, patients treated with NTLA-2002 had the greatest improvement in the functioning domain, with a mean change from baseline of −22.8±16.0 in the 25-mg group and −25.0±19.4 in the 50-mg group.

Phase 1 Update

As of February 12, 2024, the 10 patients treated with NTLA-2002 in phase 1 had a median follow-up time of 20.1 months, which represents approximately 1 year of additional follow-up after the data-cutoff date for the initial report.[17] The additional data show that the pharmacodynamic and clinical responses have been durable and ongoing, and no unexpected safety events have occurred (see the Supplementary Appendix).

Discussion

In this placebo-controlled trial of a systemically delivered, in vivo CRISPR-Cas9–based therapy, the frequency of angioedema attacks was lower among patients who received a one-time infusion of NTLA-2002 at a dose of 25 mg or 50 mg than among those who received placebo. Of 11 patients who received 50 mg of NTLA-2002, 8 were attack-free during the period from week 1 through week 16, a result that is consistent with the greater and more stable reductions achieved in total plasma kallikrein protein levels with 50 mg than with 25 mg. Among the 10 patients who received 25 mg, only 4 were attack-free.

Of the three patients who received 50 mg of NTLA-2002 and were not attack-free during the primary observation period, one (patient 2) had a reduction of 73% from baseline in the monthly attack rate. The other two patients in the 50-mg group (patients 1 and 7) had more modest, albeit clinically meaningful, reductions from baseline in the frequency of angioedema attacks. From week 1 through week 16, patient 1 had 2.5 attacks per month, an increase of 29% from baseline.

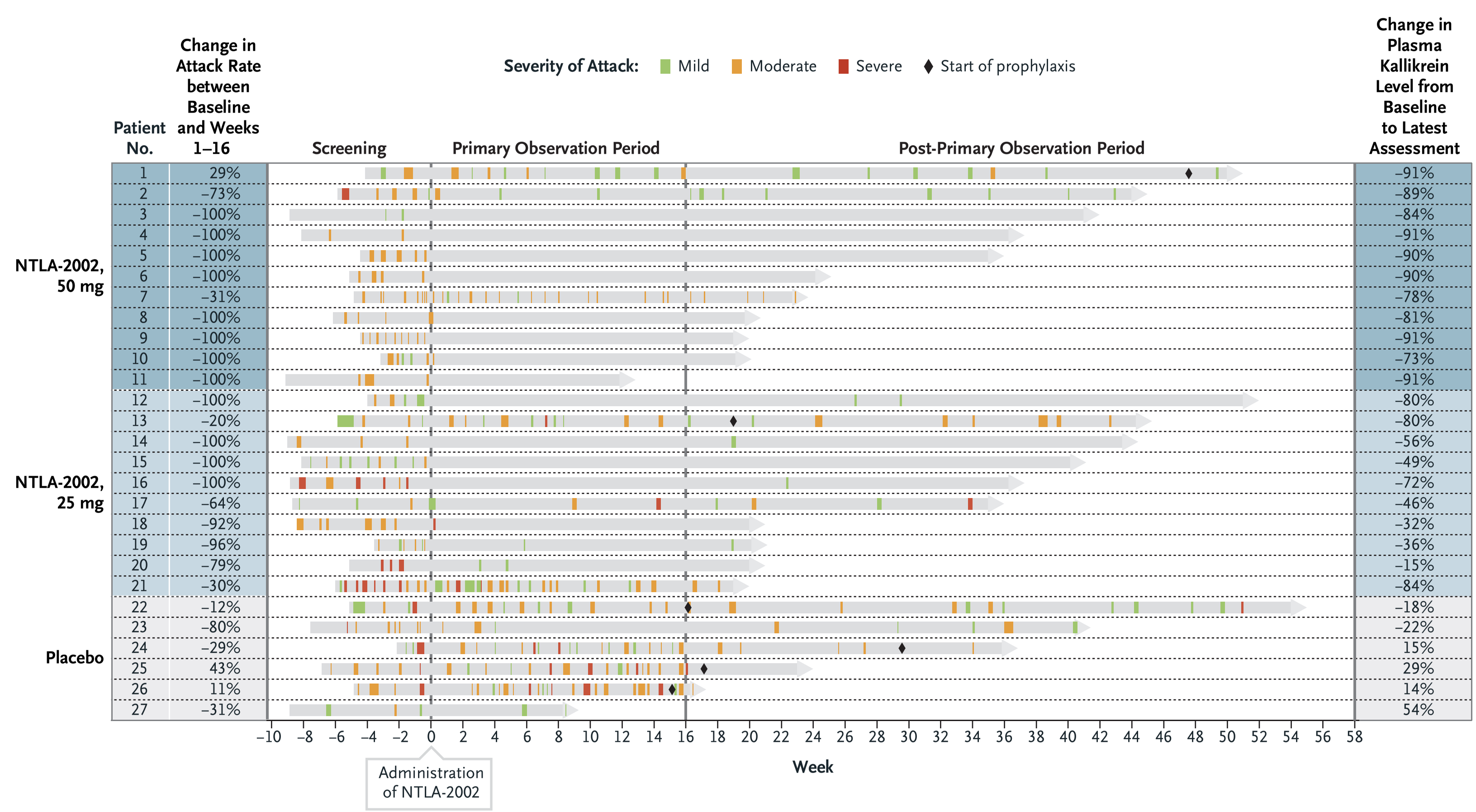


Figure 1. **Angioedema Attacks before and after Administration of NTLA-2002.**

The severity and length of angioedema attacks in individual patients during the 8-week screening period, the 16-week primary observation period, and the post–primary observation period are represented by colored bars, with the width of each bar indicating the duration of the attack. Patient data are presented from top to bottom according to the trial group and length of follow-up. Baseline is defined as the period from the date of informed consent to randomization. Gray bars represent the interval between the beginning of the screening period for the individual patient and the last assessment of that patient as of the data-cutoff date. Bars that end in arrowheads indicate that the patient is still participating in the trial. Patients were permitted to resume long-term prophylaxis after week 16.

Table 3. Adverse Events after the Administration of NTLA-2002 or Placebo.*

Event	NTLA-2002, 25 mg (N=10)	NTLA-2002, 50 mg (N=11)	NTLA-2002, Any Dose (N=21)	Placebo (N=6)
	number of patients (percent)			
Any adverse event†	10 (100)	11 (100)	21 (100)	6 (100)
Any serious adverse event	0	0	0	1 (17)‡
Adverse events				
Headache	4 (40)	4 (36)	8 (38)	1 (17)
Fatigue	3 (30)	3 (27)	6 (29)	2 (33)
Nasopharyngitis	3 (30)	3 (27)	6 (29)	2 (33)
Back pain	3 (30)	2 (18)	5 (24)	0
Upper respiratory tract infection	3 (30)	2 (18)	5 (24)	1 (17)
Cough	3 (30)	1 (9)	4 (19)	0
Infusion-related reaction	1 (10)	3 (27)	4 (19)	1 (17)
Covid-19	2 (20)	1 (9)	3 (14)	1 (17)
Ear infection	2 (20)	0	2 (10)	0
Epistaxis	0	2 (18)	2 (10)	1 (17)
Influenza-like illness	1 (10)	1 (9)	2 (10)	0
Oropharyngeal pain	1 (10)	1 (9)	2 (10)	1 (17)
Pyrexia	0	2 (18)	2 (10)	0
Sinusitis	1 (10)	1 (9)	2 (10)	0

* Adverse events that occurred in at least two patients after the administration of NTLA-2002 (at either dose) or placebo are listed according to preferred terms in the *Medical Dictionary for Regulatory Activities*, version 24.0. Covid-19 denotes coronavirus disease 2019.

† All events were grade 1 or 2 except for one grade 4 event, edema of the tongue, that occurred in a patient in the placebo group.

‡ This patient had grade 4 edema of the tongue.

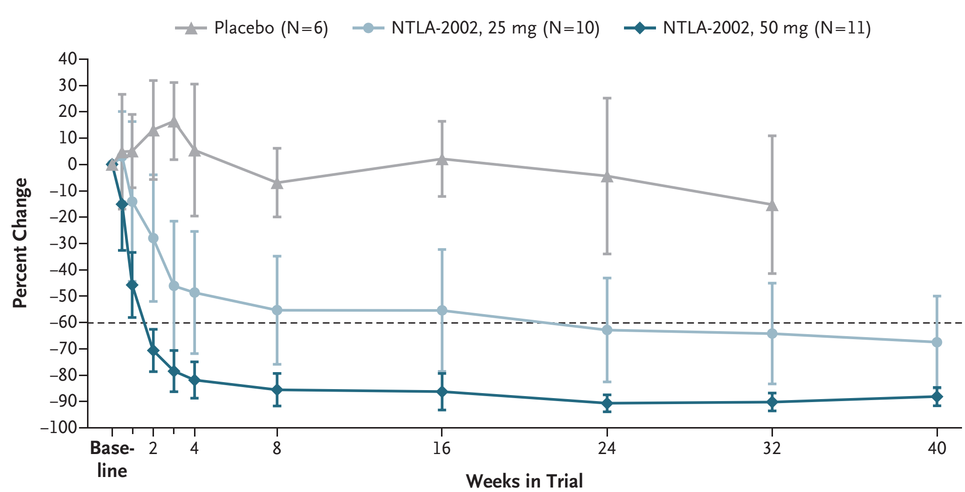
However, in the post–primary observation period during which patients remained unaware of their group assignments (week 16 until the data-cutoff point — 221 days for patient 1), patient 1 had a monthly attack rate of 0.89, a reduction of 54% from baseline, before resuming treatment with berotralstat. Patient 7 had 3.75 attacks per month from week 1 through week 16, a reduction of 31% from baseline. In the post–primary observation period (48 days for patient 7), the monthly attack rate decreased to 2.92, an overall reduction of 46% from baseline, and this patient had not resumed any long-term prophylaxis. Both patients reported having had frequent breakthrough attacks while they were taking plasma kallikrein inhibitor medications (berotralstat and lanadelumab, respectively) before participating in the trial. Both had baseline plasma kallikrein protein levels similar to those of other patients, and both had large and durable reductions in total plasma kallikrein protein levels (91% and 78%, respectively). Whether these patients would benefit further from greater reductions in plasma kallikrein protein levels or activity or whether other, unknown mechanisms underpin the remaining symptoms is unclear.

Updated data from the phase 1 portion of the trial, with a median follow-up time of 20.1 months, suggest that NTLA-2002 provides a sustained, long-term benefit. Reduced plasma kallikrein protein levels have remained stable in phase 1 patients. As of the latest follow-up, the monthly attack rate had decreased by 98% from baseline, and 8 of 10 patients have remained attack-free since the end of the primary observation period. These sustained reductions in plasma kallikrein protein levels are consistent with the findings from preclinical studies of NTLA-2002[19,20] and suggest that the effects of *KLKB1* editing have persisted through a substantial amount of hepatocyte turnover, given that a recent study showed that approximately 19% of hepatocytes are replaced each year in a healthy human liver.[21] No phase 1 patients have resumed long-term prophylaxis. In the combined analysis of phase 1–2 data, a total of 12 of 15 patients (80%) who received 50 mg of NTLA-2002 have remained attack-free for periods ranging from 12 weeks (phase 2) to approximately 19 months (phase 1), with observation ongoing.

No unexpected safety concerns arose from this phase 2 trial, and the safety of NTLA-2002 is further supported by additional follow-up data from phase 1. In phase 2, all adverse events were grade 1 or 2 except for one grade 4 serious adverse event of tongue swelling due to underlying hereditary angioedema, which occurred in a patient who received placebo. Infusion-related reactions were grade 1 or, in two patients, grade 2; these resolved without sequelae. Among a total of 31 patients who received NTLA-2002 in phases 1 and 2, there were no adverse events of grade 3 or higher, no serious adverse events, and no clinically significant shifts in coagulation variables. In one patient, an increase in the alanine aminotransferase level (grade 2, clinically nonsignificant) was observed.

Hereditary angioedema is associated with a substantial disease burden, which affects quality of life both during and between attacks. Patients who received NTLA-2002 had clinically meaningful improvements in AE-QoL scores as compared with those who received placebo.[23]

Potential limitations of this trial include the small sample size, nonstratified randomization, the short duration of follow-up, and a primary observation period of only 16 weeks. Furthermore, although the race and ethnicity of the trial population represent those of persons with hereditary angioedema in the countries in which the trial was carried out, they do not reflect the racial and ethnic groups affected by hereditary angioedema more generally. Baseline monthly attack rates and historical attack severity were well balanced among the groups, and the incidence of the disease is not known to be affected by racial or ethnic demographics,[24] which suggests that these limitations do not substantially affect the interpretation of the data.



No. at Risk

	Baseline	2	4	8	16	24	32	40
Placebo	6	6	6	5	5	4	3	
NTLA-2002, 25 mg	10	10	10	10	10	6	6	4
NTLA-2002, 50 mg	11	11	11	11	10	5	5	3

Figure 2. Change from Baseline in Total Plasma Kallikrein Protein Levels over Time According to Trial Group.

Shown is the mean percent change from baseline in the total plasma kallikrein protein level in the NTLA-2002 and placebo groups. I bars indicate standard deviations. The dashed line represents the targeted minimum reduction.[17] For postbaseline assessments, data from scheduled visits completed by at least three patients in each group are presented.

In this phase 2 placebo-controlled trial, with additional data from phase 1 follow-up, treatment with NTLA-2002 resulted in a reduction in angioedema attacks and a dose-dependent reduction in total plasma kallikrein protein levels. Furthermore, many patients treated with NTLA-2002 seemed to have a complete response to treatment, with no angioedema attacks occurring in the absence of additional treatment (prophylactic or on-demand). No new safety concerns arose, and no serious adverse events occurred in patients who received NTLA-2002. These results show the potential of a single dose of the new CRISPR-Cas9–based in vivo gene-editing therapy NTLA-2002 to be a functional cure for patients with hereditary angioedema and support continued investigation in a larger phase 3 trial.

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Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

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APPENDIX

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